| Ex:No: 1C<br>Date: 28.01.26 | Fibonacci Staircase Pattern Using Loops |
|---|---|

**Problem Description:**

A math teacher wants to print a **staircase pattern using Fibonacci numbers**. Each row of the staircase should contain Fibonacci numbers in sequence.

For example, if N = 5, the pattern should be:

```
0
0 1
0 1 1
0 1 1 2
0 1 1 2 3
```

Write an R program that takes N (number of rows) as input and prints the Fibonacci staircase using:

1. for loop
2. while loop

**Objective:**

To demonstrate the use of loops in R to generate a Fibonacci series in a structured staircase pattern, combining iterative computation with nested loops for practical pattern printing.

| Sample Input | Expected Output |
|---|---|
| Enter the number of rows: 6 | Fibonacci Staircase using for loop:<br>0<br>0 1<br>0 1 1<br>0 1 1 2<br>0 1 1 2 3<br>0 1 1 2 3 5<br>Fibonacci Staircase using while loop:<br>0<br>0 1<br>0 1 1<br>0 1 1 2<br>0 1 1 2 3<br>0 1 1 2 3 5 |

SAMPLE INPUT:

Enter the number of rows : 5

EXPECTED OUTPUT:

Fibonacci staircase using for loop:

```
0
0 1
0 1 1
0 1 1 2
0 1 1 2 3
```

Fibonacci staircase using while loop:

```
0
0 1
0 1 1
0 1 1 2
0 1 1 2 3
```

**Program:**

```
# Fibonacci Staircase Pattern Using Loops
# Function to generate nth Fibonacci number
fib <- function (n){
  if (n==1)
    return (0)
  if (n==2)
    return (1)
a <- 0
b <- 1
for (k in 3:n){
  c <- a+b
  a <- b
  b <- c
}
return(b)
}
```

```
# User input
n <- as.integer (readline ("Enter the number of rows: "))
```

```
# Using for loop
cat ("\n Fibonnaci staircase using for loop:\n")
for (i in 1:n){
  for (j in 1:i){
    cat (fib(j), " ")
  }
  cat("\n")
}
```

# Using while loop

```
cat("\n Fibonacci staircase using while loop:\n")
i<-1
while(i<=n){
  j<-1
  while(j<=i){
    cat(fib(j)," ")
    j<-j+1
  }
  cat("\n")
  i<-i+1
}
```

**Result:**

Thus, the program generates a Fibonacci staircase pattern for the given number of rows using both for and while loops. It demonstrates iterative computation of Fibonacci numbers combined with nested loops for pattern printing.